REVISION COM110

1. Given that $\int_1^{10} f(x)\,dx = 16$, $\int_{10}^{20} f(x)\,dx = 5$ and $\int_{-5}^{20} f(x)\,dx = -10$, determine the value of the following:

   (a) $\int_1^{20} f(x)\,dx$

   (b) $\int_{10}^{-5} f(x)\,dx$

   (c) $\int_{-5}^{1} f(x)\,dx$

2. Show that $y = 6 - \frac{A}{x^2}$ is a solution of the differential equation $2x\frac{dy}{dx} + 4y = 24$, where A is a constant.

3. The world population at the beginning of 1980 was 4.5 billion. Assuming that the population continued to grow at the rate of approximately 2% per year, the population growth can be modelled by the differential equation $\frac{dQ}{dt} = 0.02Q$, where $Q(t)$ represents the population (in billions) at time (in years, with $t = 0$ corresponding to 1980).

   (a) Prove that the general solution to this differential equation is $Q(t) = 4.5e^{0.02t}$.

   (b) Using the result from (a), what will be the world population at the beginning of 2019. Round your final answer to four decimal places (billions).

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

   (a) Prove that $P(t) = \frac{1}{1 + Ae^{-kt}}$.

   (b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

1. Given that $\int_1^{10} f(x)\,dx = 16$, $\int_{10}^{20} f(x)\,dx = 5$ and $\int_{-5}^{20} f(x)\,dx = -10$, determine the value of the following:

(a) $\int_1^{20} f(x)\,dx$

(b) $\int_{10}^{-5} f(x)\,dx$

(c) $\int_{-5}^{1} f(x)\,dx$

a) $\int_1^{20} f(x)\,dx = \int_1^{10} f(x)\,dx + \int_{10}^{20} f(x)\,dx = 16 + 5 = 21$

b) $\int_{10}^{-5} f(x)\,dx = \int_{10}^{20} f(x) + \int_{20}^{-5} f(x)\,dx$

$= 5 - (-10) = 15$

c) $\int_{-5}^{1} f(x)\,dx = \int_{-5}^{10} f(x)\,dx + \int_{10}^{1} f(x)\,dx$

$= -15 - 16 = -31$

2. Show that $y = 6 - \frac{A}{x^2}$ is a solution of the differential equation $2x\frac{dy}{dx} + 4y = 24$, where A is a constant.

$y' = +2Ax^{-3}$

$2x \cdot y' + 4y = 2x(+2Ax^{-3}) + 4(6 - \frac{A}{x^2})$

$= 4Ax^{-2} + 24 - \frac{4A}{x^2}$

$= 24$

LHS of DE = RHS of DE

$\therefore \quad y = 6 - \frac{A}{x^2}$ ...

3. The world population at the beginning of 1980 was 4.5 billion. Assuming that the population continued to grow at the rate of approximately 2% per year, the population growth can be modelled by the differential equation $\frac{dQ}{dt} = 0.02Q$, where $Q(t)$ represents the population (in billions) at time (in years, with $t = 0$ corresponding to 1980).

(a) Prove that the general solution to this differential equation is $Q(t) = 4.5e^{0.02t}$.

(b) Using the result from (a), what will be the world population at the beginning of 2019. Round your final answer to four decimal places (billions).

a) $\frac{dQ}{dt} = 0.02Q$

$\frac{1}{Q}\,dQ = 0.02\,dt$

$\int \frac{1}{Q}\,dQ = 0.02 \int dt$

$\ln|Q| = 0.02t + C$

$|Q| = e^{0.02t + C}$

$Q = Ae^{0.02t}$ where $A = \pm e^{C}$

$Q(0) = A = 4.5$

$\therefore Q(t) = 4.5e^{0.02t}$

b) $2019 - 1980 = 39$

$Q(39) = 4.5e^{0.02 \times 39} = 9.8166$ billion

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

(a) Prove that $P(t) = \frac{1}{1+Ae^{-kt}}$.

(b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

a) $\frac{1}{P(1-P)}\,dP = k\,dt$

$\int \frac{A}{P} + \frac{B}{1-P}\,dP = k\int dt$

$A(1-P) + BP = 1$

① when $P = 1$

$+B = 1 \Rightarrow B = +1$

② when $P = 0$

$A = 1$

$\int \frac{1}{P} + \frac{1}{1-P}\,dP = kt + C$

$\ln|P| - \ln|1-P| = kt + C$

$\ln\left|\frac{P}{1-P}\right| = kt + C$

$\ln\left|\frac{1-P}{P}\right| = -kt - C$

$\left|\frac{1-P}{P}\right| = e^{-kt-C}$

$\frac{1}{P} - 1 = Ae^{-kt}$, where $A = 1\,e^{-C}$

$\frac{1}{P} = A\cdot e^{-kt} + 1 \Rightarrow P = \frac{1}{Ae^{-kt}+1}$

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

(a) Prove that $P(t) = \frac{1}{1+Ae^{-kt}}$.

$P = \frac{80}{1000}$

(b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

$P = 0.9$

b) let 8am be T_0

$$P(t) = \frac{1}{1+Ae^{-0.6106t}}$$

$$P(0) = \frac{1}{1+A} = \frac{80}{1000} \Rightarrow A = 11.5$$

$$\therefore P(t) = \frac{1}{1+11.5e^{-0.6106t}}$$

$t = 3.36$ pm

$$0.9 = \frac{1}{1+11.5e^{-0.6106t}}$$

$$11.5e^{-0.6106t} = \frac{1}{9}$$

$$e^{-0.6106t} = \frac{2}{207}$$

$$-0.6106t = \ln\left(\frac{2}{207}\right)$$

$t = 7.60\ h = 7\ h\ 36\ min$

5. Solve the ordinary differential equation $y''+3y'+2y=x^2+5$ by the method of undetermined coefficients.

6. Solve the initial value problem $y''-4y'-12y=3e^{5t}, y(0)=\frac{18}{7}, y'(0)=\frac{-1}{7}$.

7. Evaluate the following indefinite integral:
   (a) $\int x^5+2x\,dx$
   (b) $\int \frac{3}{2x}\,dx$
   (c) $\int \frac{x^5+x+5}{x^2}\,dx$
   (d) $\int \sin^2(x)\cos^5(x)\,dx$

8. Evaluate the definite integral $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\cos x}{\sin^2 x}\,dx$. Round your final answer to four decimal places.

9. Use an appropriate substitution to solve the indefinite integral $\int 2x^2 e^{x^3}\,dx$.

10. Use an integration by parts to solve the indefinite integral $\int 2x\cos(5x)\,dx$.

11. Consider a rational function, $f(x)=\frac{x+11}{x^2+x-6}$.
    (a) Perform partial fraction decomposition on the given rational function.
    (b) Using the partial fraction decomposition obtained in (a), evaluate the definite integral $\int_5^6 f(x)\,dx$. Round your final answer to four decimal places.

12. Use a trigonometric substitution to solve the indefinite integral $\int \frac{\sqrt{16-x^2}}{x^2}\,dx$.

5. Solve the ordinary differential equation $y''+3y'+2y=x^2+5$ by the method of undetermined coefficients.

① $y_c = e^{rx}$

$y_c' = re^{rx}$

$y_c'' = r^2 e^{rx}$

$r^2 e^{rx} + 3re^{rx} + 2e^{rx} = 0$

$(r^2+3r+2)e^{rx} = 0$

$e^{rx} = 0$ (impossible) or $r^2+3r+2=0$

$(r+2)(r+1)=0$

$r=-2$ or $r=-1$

$\therefore y_c = c_1 e^{-2x} + c_2 e^{-x}$

② $y_p = Ax^2 + Bx + C$

$y_p' = 2Ax + B$

$y_p'' = 2A$

$2A + 6Ax + 3B + 2Ax^2 + 2Bx + 2C = x^2 + 5$

$2Ax^2 + (6A+2B)x + 2A + 3B + 2C = x^2 + 5$

$2A = 1 \Rightarrow A = \frac{1}{2}$

$6A + 2B = 0 \Rightarrow B = -\frac{3}{2}$

$2A + 3B + 2C = 5 \Rightarrow C = \frac{17}{4}$

$y_p = \frac{1}{2}x^2 - \frac{3}{2}x + \frac{17}{4}$

$y = y_c + y_p$

$= c_1 \cdot e^{2x} + c_2 \cdot e^{-x} + \frac{1}{2}x^2 - \frac{3}{2}x + \frac{17}{4}$

Summary: Solutions of $ay'' + by' + c = 0$

Roots of $ar^2 + br + c = 0$	General solution
r_1, r_2 real and distinct	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$r_1 = r_2 = r$	$y = c_1 e^{rx} + c_2 x e^{rx}$
r_1, r_2 complex: $\alpha \pm i\beta$	$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

6. Solve the initial value problem $y''-4y'-12y=3e^{5t}, y(0)=\frac{18}{7}, y'(0)=\frac{-1}{7}$.

let $y_c = e^{rx}$

$y_c' = re^{rx}$

$y_c'' = r^2e^{rx}$

y_c 和 y_p 結合一起

代入原式

$r^2e^{rx} - 4re^{rx} - 12e^{rx} = 0$

$(r^2 - 4r - 12)e^{rx} = 0$

$e^{rx} = 0$ 或 $r^2 - 4r - 12 = 0$

$(r-6)(r+2) = 0$

$r = 6$ 或 $r = -2$

$\therefore y_c = c_1e^{6x} + c_2e^{-2x}$

let $y_p = Ae^{5t}$

$y_p' = 5A \cdot e^{5t}$

$y_p'' = 25Ae^{5t}$

$25Ae^{5t} - 20Ae^{5t} - 12Ae^{5t} = 3e^{5t}$

$-7Ae^{5t} = 3e^{5t}$

$A = -\frac{3}{7}$

$\therefore y_p = -\frac{3}{7}e^{5t}$

$$y = y_c + y_p$$

$$= c_1 e^{6x} + c_2 e^{-2x} - \frac{3}{7} e^{5t}$$

$$y(0) = c_1 + c_2 - \frac{3}{7} = \frac{18}{7}$$

$$c_1 + c_2 = 3$$

$$y' = 6c_1 e^{6x} - 2c_2 e^{-2x} - \frac{15}{7} e^{5t}$$

$$y'(0) = 6c_1 - 2c_2 - \frac{15}{7} = -\frac{1}{7}$$

$$6c_1 - 2c_2 = 2$$

$$\Rightarrow c_1 = 1$$

$$c_2 = 2$$

$$\therefore y = e^{6t} + 2e^{-2t} - \frac{3}{7} e^{5t}$$

7. Evaluate the following indefinite integral:

(a) $\int x^5 + 2x\,dx$

$= \frac{x^6}{6} + x^2 + C$

(b) $\int \frac{3}{2x}\,dx$

$= \frac{3}{2} \ln|x| + C$

(c) $\int \frac{x^5 + x + 5}{x^2}\,dx$

$= \int x^3 + \frac{1}{x} + \frac{5}{x^2}\,dx$

$= \frac{x^4}{4} + \ln|x| - 5\frac{1}{x} + C$

(d) $\int \sin^2(x)\cos^5(x)\,dx$

$= \int \sin^2(x) \cdot \cos^2 x \cdot \cos^2 x \cdot \cos x\,dx$

$= \int \sin^2 x \cdot (1 - \sin^2 x)(1 - \sin^2 x)\cos x\,dx$

let $u = \sin x$; $dx = \frac{du}{\cos x}$

$= \int u^2 (1 - u^2)^2\,du$

$= \int u^2 (1 - 2u^2 + u^4)\,du$

$= \int u^2 - 2u^4 + u^6\,du = \frac{u^3}{3} - \frac{2}{5}u^5 + \frac{1}{7}u^7 + C = \frac{\sin^3 x}{3} - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C$

8. Evaluate the definite integral $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\cos x}{\sin^2 x}\,dx$. Round your final answer to four decimal places.

Let $u =$

$\cot x \quad -\csc^2 x$

$\csc x \quad -\cot x \csc x$

$\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x}$

$\frac{1}{\sin^2 x}$

$\int \cot x \cdot \csc x \, dx$

$-\left[\csc x\right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}}$

12. Use a trigonometric substitution to solve the indefinite integral $\int \frac{\sqrt{16-x^2}}{x^2}\,dx$.

let $x = 4\sin\theta$; $dx = 4\cos\theta\, d\theta$

$\sin\theta = \frac{x}{4}$

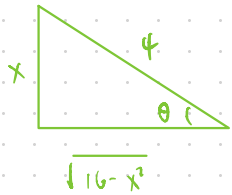
$$= \int \frac{\sqrt{16 - 16\sin^2\theta}}{16\sin^2\theta} \cdot 4\cos\theta\, d\theta = \int \frac{16\cos^2\theta}{16\sin^2\theta}\, d\theta$$

$$= \int \cot^2\theta\, d\theta$$

$$= \int (\csc^2\theta - 1)\, d\theta$$

$$= -\cot\theta - \theta + C$$

$$= -\frac{\sqrt{16-x^2}}{x} - \sin^{-1}\left(\frac{x}{4}\right) + C$$



$\cot\theta = \frac{1}{\tan\theta} = \frac{\sqrt{16-x^2}}{x}$

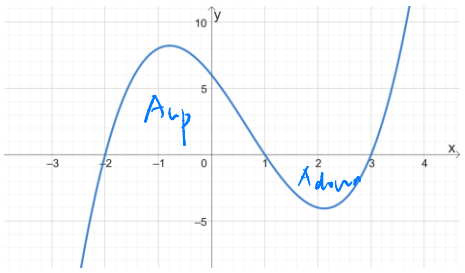
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Expression	Substitution	Domain	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$0 \le \theta \le \pi,\ \theta \ne \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

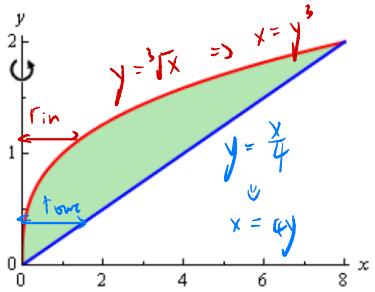
13. Use a trigonometric substitution to solve the definite integral $\int_{\frac{-4}{5}}^{\frac{-2}{5}} \frac{\sqrt{25x^2-4}}{x}\,dx$.

Round your final answer to four decimal places.

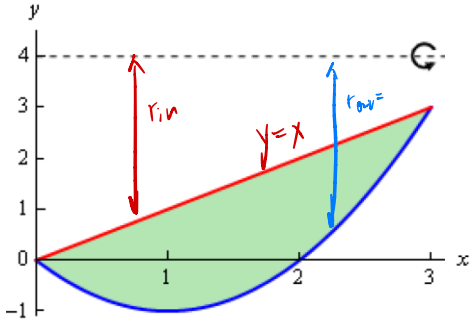
14. Figure below shows a graph of the functions $f(x)=x^3-2x^2-5x+6$. Compute the area of the region bounded by the $x-$axis, $y=f(x)$, and the lines $x=-2$ and $x=3$. Round your final answer to four decimal places.



15. Figure below shows the region enclosed by the curves $y=\sqrt[3]{x}$ and $y=\frac{x}{4}$ that lies in the first quadrant. Find the volume of the solid obtained by rotating the portion of the region about the *y-axis* through Washer method.



16. Figure below shows the region enclosed by the curves $y=x^2-2x$ and $y=x$. Find the volume of the solid obtained by rotating the portion of the region about the $y=4$ through Washer method.



13. Use a trigonometric substitution to solve the definite integral $\int_{\frac{-4}{5}}^{\frac{-2}{5}} \frac{\sqrt{25x^2-4}}{x}\,dx$.

Round your final answer to four decimal places.

let $5x = 2\sec\theta$

$x = \frac{2}{5}\sec\theta$; $dx = \frac{2}{5}\sec\theta\tan\theta\,d\theta$

$\sec\theta = \frac{5}{2}x$

$\cos\theta = \frac{2}{5x}$

$\theta = \cos^{-1}\left(\frac{2}{5x}\right)$

when $x = -\frac{2}{5}$, $\theta = \cos^{-1}\left(\frac{2}{5\times(-\frac{2}{5})}\right) = \cos^{-1}\left(\frac{2}{-2}\right) = \cos^{-1}(-1) = \pi$

$x = -\frac{4}{5}$, $\theta = \cos^{-1}\left(\frac{2}{5\times(-\frac{4}{5})}\right) = \cos^{-1}\left(\frac{1}{2}\right) = \frac{2}{3}\pi$

$$= \int_{\frac{2\pi}{3}}^{\pi} \frac{\sqrt{4(\sec^2\theta-1)}}{\frac{2}{5}\sec\theta}\cdot\frac{2}{5}\sec\theta\tan\theta\,d\theta$$

$$= \int_{\frac{2\pi}{3}}^{\pi} \frac{2\tan\theta}{\cancel{\frac{2}{5}\sec\theta}}\cdot\cancel{\frac{2}{5}\sec\theta}\tan\theta\,d\theta = 2\int_{\frac{2\pi}{3}}^{\pi}\tan^2\theta\,d\theta$$

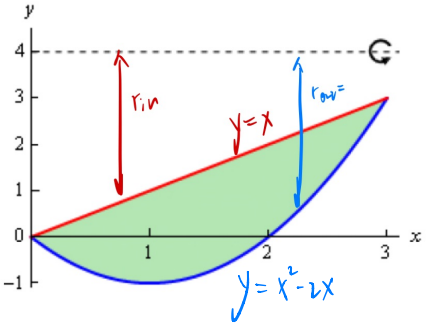
$$= 2\int_{\frac{2\pi}{3}}^{\pi}(\sec^2\theta-1)\,d\theta$$

$$= 2\left[\tan\theta-\theta\right]_{\frac{2\pi}{3}}^{\pi}$$

$$= 2\left[\tan\pi-\pi-\left(\tan\frac{2\pi}{3}-\frac{2\pi}{3}\right)\right]$$

$$= 1.5697$$

16. Figure below shows the region enclosed by the curves $y = x^2 - 2x$ and $y = x$. Find the volume of the solid obtained by rotating the portion of the region about the $y = 4$ through Washer method.



$$A = \int_0^3 \pi r_{out}^2 - \pi r_{in}^2 \, dx$$

$$= \int_0^3 \pi \left[4 - (x^2 - 2x)\right]^2 - \pi (4 - x)^2 \, dx$$

$$= \int_0^3 \pi \left[16 - 8(x^2 - 2x) + (x^2 - 2x)^2\right] - \pi (16 - 8x + x^2) \, dx$$

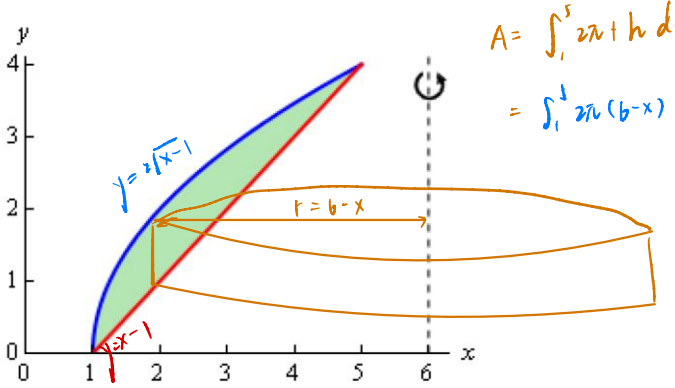
$$= \int_0^3 \pi \left[16 - 8x^2 + 16x + x^4 - 4x^3 + 4x^2\right] - \pi (16 - 8x + x^2) \, dx$$

$$= \int_0^3 \pi (16 - 4x^2 + 16x + x^4 - 4x^3] - \pi (16 - 8x + x^2) \, dx$$

$$= \int_0^3 \pi (16 + 16x - 4x^2 - 4x^3 + x^4 - 16 + 8x - x^2) \, dx$$

$$= \int_0^3 \pi (24x - 5x^2 - 4x^3 + x^4) \, dx$$

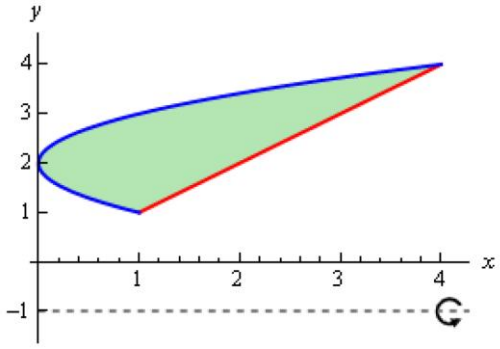
17. Figure below shows the region enclosed by the curves $y = 2\sqrt{x-1}$ and $y = x-1$. Find the volume of the solid obtained by rotating the portion of the region about the $x = 6$ through Cylindrical Shell method.



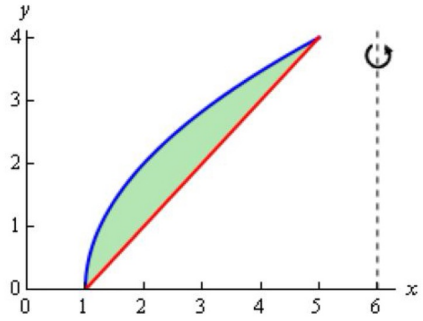
$A = \int_1^5 2\pi \cdot h \, dx$

$= \int_1^5 2\pi (6-x)\left[2\sqrt{x-1} - (x-1)\right] dx$

18. Figure below shows the region enclosed by the curves $x = (y-2)^2$ and $y = x$. Find the volume of the solid obtained by rotating the portion of the region about the $y = -1$ through Cylindrical Shell method.



17. Figure below shows the region enclosed by the curves $y = 2\sqrt{x-1}$ and $y = x-1$. Find the volume of the solid obtained by rotating the portion of the region about the $y = 6$ through Cylindrical Shell method.



$A = \int_1^5 2\pi \cdot h \, dx$

$= \int_1^5 2\pi (6-x)\left[2\sqrt{x-1} - (x-1)\right] dx$

$= 2\pi \int_1^5 12\sqrt{x-1} - 6(x-1) - 2x\sqrt{x-1} + x(x-1)\, dx$

$= 2\pi \int_1^5 x^2 - 7x + 6 + 12\sqrt{x-1} - 2x\sqrt{x-1}\, dx$

let $u = x-1$

$= 2\pi \left[\frac{x^3}{3} - \frac{7}{2}x^2 + 6x\right]_1^5 + 2\pi \int_1^5 12\sqrt{x-1} - 2x\sqrt{x-1}\, dx$

let $u = x-1$; $du = dx$

$x = u+1$

when $x = 1$, $u = 0$

$x = 5$, $u = 4$

$= 2\pi\left[\frac{5^3}{3} - \frac{7}{2}\cdot 5^2 + 6\times 5 - \left(\frac{1}{3} - \frac{7}{2} + 6\right)\right] + 2\pi \int_0^4 12\sqrt{u} - 2(u+1)\sqrt{u}\, du$

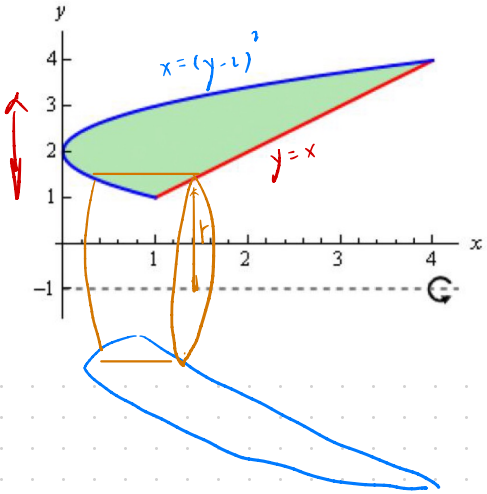
$= 2\pi \times \frac{56}{3} + 2\pi \times 12 \times \frac{2}{3}\left[u^{\frac{3}{2}}\right]_0^4 - 4\pi \int_0^4 u^{\frac{3}{2}} + u^{\frac{1}{2}}\, du$

$= \frac{112}{3}\pi + 16\pi\left(4^{\frac{3}{2}} - 0\right) - 4\pi\left[\frac{2}{5}u^{\frac{5}{2}} + \frac{2}{3}u^{\frac{3}{2}}\right]_0^4$

$= \frac{112}{3}\pi + 128\pi - 4\pi\left(\frac{2}{5}\cdot 4^{\frac{5}{2}} + \frac{2}{3}\cdot 4^{\frac{3}{2}}\right)$

$= \frac{112}{3}\pi + 128\pi - \frac{1088}{15}\pi = \frac{856}{15}\pi$ ✗ $\frac{272}{15}\pi$

18. Figure below shows the region enclosed by the curves $x=(y-2)^2$ and $y=x$. Find the volume of the solid obtained by rotating the portion of the region about the $y=-1$ through Cylindrical Shell method.



$$A = \int_0^4 2\pi \cdot h \, dy$$

$$= \int_0^4 2\pi \cdot (y+1) \cdot [y - (y-2)^2] \, dy$$

$$= \int_0^4 2\pi (y+1)[y - (y^2 - 4y + 4)] \, dy$$

$$= \int_0^4 2\pi (y+1)(-y^2 + 5y - 4) \, dy$$

$$= 2\pi \int_0^4 (-y^3 + 5y^2 - 4y - y^2 + 5y - 4) \, dy$$

$$= 2\pi \int_0^4 (-y^3 + 4y^2 + y - 4) \, dy$$

$$= 2\pi \left[-\frac{y^4}{4} + \frac{4}{3}y^3 + \frac{y^2}{2} - 4y \right]_0^4 = 2\pi \left[-\frac{4^4}{4} + \frac{4}{3} \cdot 4^3 + \frac{4^2}{2} - 4 \times 4 \right]$$

$$\left(-\frac{1}{4} + \frac{4}{3} + \frac{1}{2} - 4\right)$$

$$= \frac{80}{3}\pi \times \frac{63}{2}\pi$$